MILITARY SPECIFICATION SHEET

SWITCHES, PUSH, 10 AMPERES OR LOW LEVEL, DUSTTIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-8805.

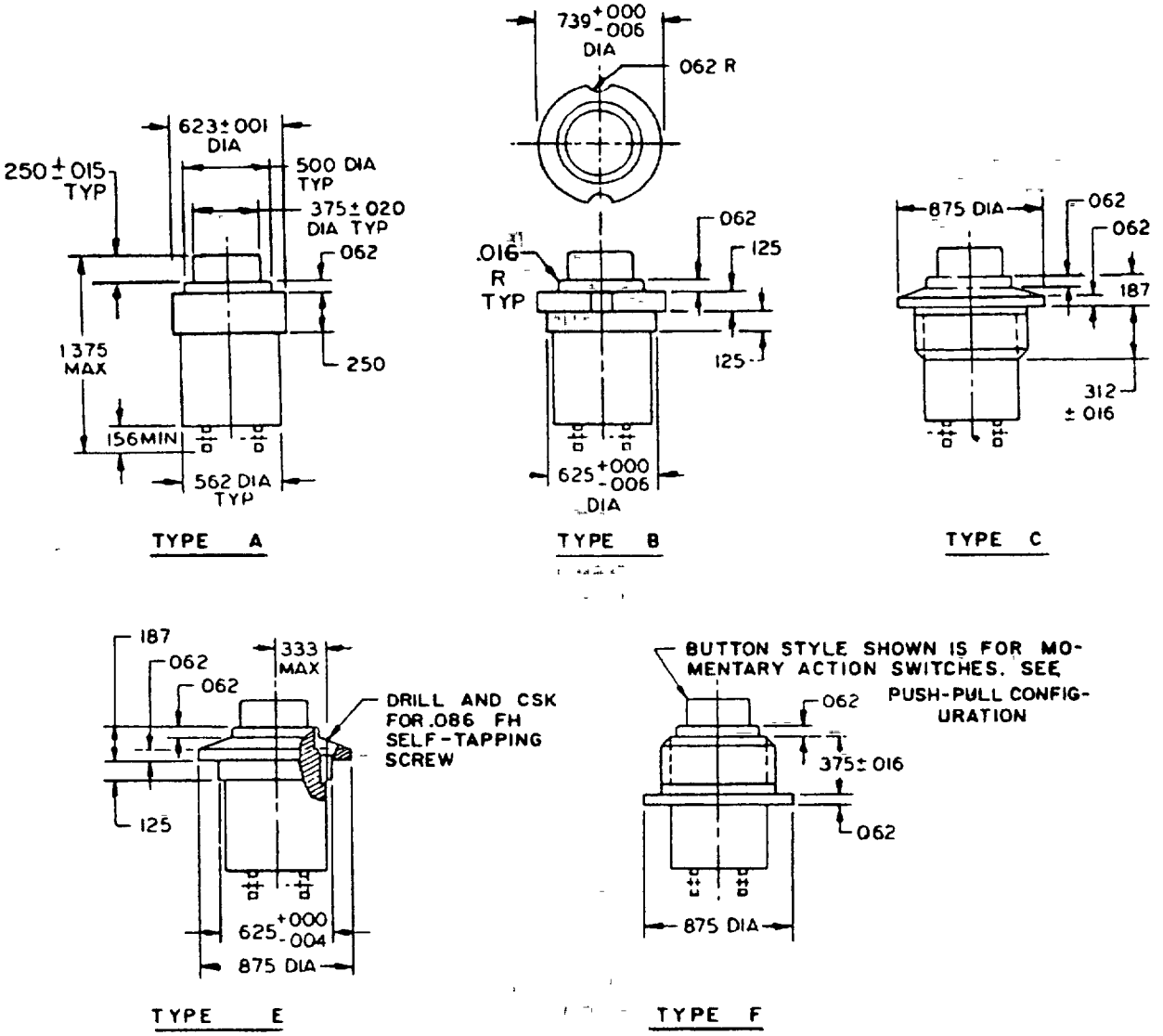
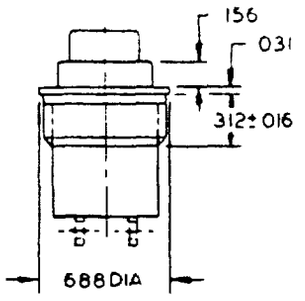


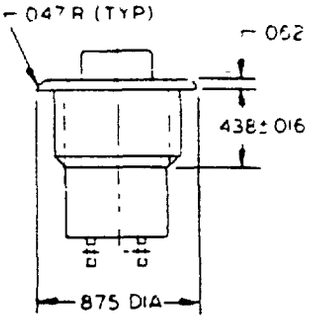
FIGURE 1 Configuration and dimensions.

Ⓗ denotes changes

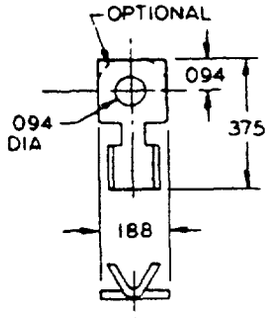
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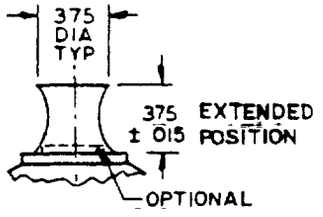
TYPE G



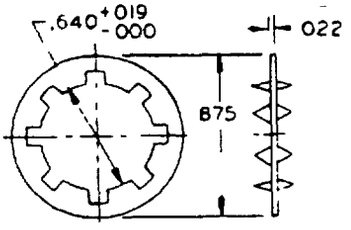
TYPE H



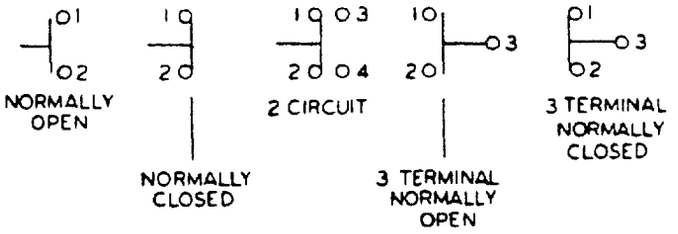
SOLDER TERMINAL LUG
ENLARGED



PUSH – PULL
CONFIGURATION
SEE NOTE 7



LOCKWASHER, MTG DETAIL
SEE NOTE 6



CIRCUIT SCHEMATIC
SEE NOTE 9

Inches	mm	Inches	mm
001	0 03	188	4 78
004	0 10	250	6 35
006	0.15	312	7 92
015	0.38	333	8 46
016	0 41	375	9 53
019	0 48	438	11 13
020	0 51	500	12 70
022	0 56	562	14 27
031	0 79	623	15 82
047	1 19	625	15 88
062	1 57	.640	16 26
094	2 39	688	17 48
125	3 18	739	18 77
156	3 96	.875	22 23
187	4 75	1.375	34 93

NOTES
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
3. Mounting thread .625-24 UNEF-2A applies to type C, F, G, and H. All threads to be within .10 (2.54 mm) of shoulder.
4. All terminal threads are 086-56UNC-2B.
5. Hardware for each terminal includes one screw, one terminal lug, and one lockwasher.
(H) 6. Types C, F, G, and H shall be furnished with one each mounting nut MS25082-B11 and lockwasher. Lockwasher may be replaced by MS35333-138.
(H) 7. Overall height of switch with push-pull activator is 1.530 (38.86 mm) maximum in extended position.
8. Metric equivalents are given for general information only.
(H) 9. For push-pull configuration, circuit schematic is shown with actuator in extended position.

FIGURE 1. Configuration and dimensions - Continued.

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REQUIREMENTS:

Dimensions and configuration. See figure 1.

Enclosure design: 2 (dusttight).

Temperature characteristic: 1 (-55°C to +85°C).

Shock type: H for momentary type only (no contact opening or closing in excess of 20 milliseconds). M for other units.

(H) Sinusoidal vibration grade: 2 (10 to 2,000 Hz).

Weight: .050 pound maximum.

(H) Solderability: Applicable to screw tabs and solder lugs.

Operating characteristics:

Total plunger travel: .085 ±.015 inch.

Operating force:

Momentary: 4 ±1 pounds.

Push-pull: 0.5 to 2 pounds.

Releasing force: 1 pound minimum, momentary.

Terminal strength torque test: Not applicable.

Strength of actuator and stop: 75 pounds for 3 terminal circuitry; 40 pounds for all others.

Material.

Case: Aluminum alloy, 2011-T3.

(H) Plunger cap: In accordance with ASTM-4066-82. May be flame retardant glass filled nylon which is self extinguishing when tested in accordance with method 2021 of FED-STD-406.

Body: Flame resistant diallyl phthalate which meets the requirements of MIL-M-14.

(H) Finish: The switch case shall be anodized in accordance with MIL-A-8625, type II. All metal surfaces which are exposed to the view of the operator shall be black anodized.

Dielectric withstanding voltage:

Sea level: 1,000 V rms.

After electrical endurance: 200 V rms.

Intermediate current: 1/

MS25089-1, MS25089-2, and MS25089-3: 200,000 cycles.

MS25089-4 and MS25089-5: 50,000 cycles.

1/ Prior to test, units preconditioned at 85°C for 48 hours.

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Mechanical endurance (10 to 15 cpm)

MS25089-1, MS25089-2, and MS25089-3

At room temperature. 200,000 cycles.

At -55°C and +85°C. 100,000 cycles each.

MS25089-4 and MS25089-5. 50,000 cycles.

Electrical endurance: 25,000 cycles at sea level, 5,000 cycles at 50,000 feet.

Electrical ratings.

Center terminal ratings. 3 amperes maximum.

Other than center terminals: See table I.

Ⓗ TABLE I. Electrical ratings, power circuit switches 1/

Load	Sea level		50,000 feet
	28 V dc	115/V ac, 60/400 Hz	28 V dc
	amperes	amperes	amperes
Resistive - - -	10	10	---
Inductive - - -	5	5	5
Lamp- - - - - -	3	3	---

Ⓗ 1/ Center terminal rating 3 amperes maximum.

Sand and dust: Applicable

Low level circuit. Applicable when specified (see table II) 25,000 cycles.

Ⓗ Electronic logic circuit (5.0 V dc, .010 ampere). Applicable when specified (see table II). 25,000 cycles at an actuation rate of 120 cycles per minute maximum with no "stick", or "misses", allowed when tested in accordance with ANSI/EIA RS448, method 17 as follows.

Test condition.

Each switch contact shall be tested using a 5.0 ±0.5 V dc, 10 ±1 mA resistive load. During each closure of the contacts, the voltage drop across the switch terminals shall be monitored for a duration of no less than 50 percent of each contact static closure. The switch contacts need not be monitored until 10 milliseconds after the initial contact closure to exclude any contact bounce. During each opening of the contacts, the voltage drop across the switch terminals shall be monitored for a duration of no less than 50 percent of each contact opening.

A voltage of 2.1 volts or greater across the switch terminals shall constitute a contact "miss" (failure to properly close the circuit). A voltage drop of less than 90 percent of the open-circuit voltage shall constitute a contact "stick" (failure to properly open the circuit).

The monitoring device shall either record the number of contact closures at which "sticks" and/or "misses" occur, or discontinue the test when "sticks" and/or "misses" occur.

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Operating temperature

a. 25 percent of the test cycle at -55°C specified.

b. 25 percent of the test cycle at room ambient conditions.

c. 50 percent of the test cycle at +85°C.

Qualification:

Group submission: See table III.

Group A inspection: When low level circuit applies, switches shall not be subjected to loads greater than 100 milliamperes at an open circuit voltage of 6 volts maximum dc or peak ac.

Ⓗ Part number: See table II.

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TABLE II. Part numbers and characteristics.

1/ Part no MS25089-																						
Momentary action switches																						
Type A			Type B			Type C			Type E			Type F			Type G			Type H			Switch circuit
Button color			Button color			Button color			Button color			Button color			Button color			Button color			
Black	Red	White	Black	Red	White	Black	Red	White	Black	Red	White	Black	Red	White	Black	Red	White	Black	Red	White	
1A	1AR	1AW	1B	1BR	1BW	1C	1CR	1CW	1E	1ER	1EW	1F	1FR	1FW	1G	1GR	1GW	1H	1HR	1HW	NO
2A	2AR	2AW	2B	2BR	2BW	2C	2CR	2CW	2E	2ER	2EW	2F	2FR	2FW	2G	2GR	2GW	2H	2HR	2HW	NC
3A	3AR	3AW	3B	3BR	3BW	3C	3CR	3CW	3E	3ER	3EW	3F	3FR	3FW	3G	3GR	3GW	3H	3HR	3HW	2 circuit
4A	4AR	4AW	4B	4BR	4BW	4C	4CR	4CW	4E	4ER	4EW	4F	4FR	4FW	4G	4GR	4GW	4H	4HR	4HW	3 term NO
5A	5AR	5AW	5B	5BR	5BW	5C	5CR	5CW	5E	5ER	5EW	5F	5FR	5FW	5G	5GR	5GW	5H	5HR	5HW	3 term NC
Push-pull switches																						
None			None			1C-1	1CR-1	1CW-1	None			1F-1	1FR-1	1FW-1	1G-1	1GR-1	1GW-1	1H-1	1HR-1	1HW-1	NO
						2C-1	2CR-1	2CW-1				2F-1	2FR-1	2FW-1	2G-1	2GR-1	2GW-1	2H-1	2HR-1	2HW-1	NC
						3C-1	3CR-1	3CW-1				3F-1	3FR-1	3FW-1	3G-1	3GR-1	3GW-1	3H-1	3HR-1	3HW-1	2 circuit
						4C-1	4CR-1	4CW-1				4F-1	4FR-1	4FW-1	4G-1	4GR-1	4GW-1	4H-1	4HR-1	4HW-1	3 term NO
						5C-1	5CR-1	5CW-1				5F-1	5FR-1	5FW-1	5G-1	5GR-1	5GW-1	5H-1	5HR-1	5HW-1	3 term NC

Ⓗ 1/ Low level circuit and electronic logic circuit does not apply for dash numbers shown. To specify switches with low level circuit and electronic logic circuit applicable, add the letter "L" to the dash numbers

Examples of part numbers:

MS25089-4C = Switch, Push, Dusttight, 3 Terminal (NO), Type C, Momentary, Black Button.
MS25089-4CR-1 = Switch, Push, Dusttight, 3 Terminal (NO), Type C, Push-Pull, Red Button.

MS25089-4CL
MS25089-4CR-1L
} As above, except low level circuit and electronic logic circuit applies. Power circuit electrical ratings do not apply.

TABLE III. Qualification inspection.

Examination or test	Basic switch	Other switch Ⓗ samples	Extent of approval	
Qualification 1/ inspection table of MIL-S-8805	Required sample units of any 2 circuit configurations 2/	Required number of sample units of 3 terminal NO or 3 terminal NC (groups VII and X tests, qualification inspection table, MIL-S-8805 plus 2 additional sample units shall be subjected to electronic logic circuit test and operating characteristics) 2/	All	In addition, 2 sample units of each of the other configurations shown on this specification sheet shall be submitted. Visual and mechanical examination.

1/ Electrical endurance tests shall be performed at 28 V dc and 115 V ac, 60 Hz only

Ⓗ 2/ One-half of sample units shall be momentary and one-half shall be push-pull

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, EA
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, AV, MI
Navy - AS, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5930-1313-1)

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